Analysis of Crime and Unemployment Time Series Post-Pandemic

1 Methods

Due to the seasonal nature of crime in Detroit in the initial data exploration, using the seasonal extension of ARMA (auto-regressive moving average) models, SARIMA was reasonable. Here, we consider SARIMA and SARIMAX. (Additional description here depending on time plot and ACF for why this is appropriate)

Model Consideration

Part I: SARIMA vs SARIMA

The time series for crime in Detroit diverged during the Covid-19 pandemic. Thus, an appropriate model for this data would account for pre-Covid, post-Covid, and during Covid seasons. There are two approaches to consider this: (1) model SARIMAX with a dummy variable for these time periods, and (2) fit 3 separate SARIMA models for the three periods. Here, data is categorized as pre, during, post Covid with the following identifiers. Let $\mathcal{T}$ denote the time series data for crime.

The indicator variables for the SARIMAX model are

$$D_{\text{during},t} = I(03/01/2020 \leq t \leq 07/01/2022), \tag{1}$$

$$D_{\text{post},t} = I(t > 07/01/2022). \tag{2}$$

where "pre" is the reference category.

The equation for SARIMAX with constant mean and the covid indicator variables:

$$\phi(B)\Phi(B^{12})\Big(Y_t - \mu - \beta_1 D_{\text{during},t} - \beta_2 D_{\text{post},t}\Big) = \theta(B)\Theta(B^{12})\varepsilon_t. \tag{3}$$

where μ is the mean of the equation for crime before covid ("pre"). While the mean can be different across the three time periods, this SARIMA model does not allow for different dynamics (ARMA, Seasonal ARMA) terms.

(2)

In order to fit SARIMA on the three subsets of crime, 3 models are necessary for each time period. The following equations detail the form of the pre-covid, during-covid, and post-covid seasons:

$$\phi_{\text{pre}}(B)\Phi_{\text{pre}}(B^{12})(Y_t - \mu_{\text{pre}}) = \theta_{\text{pre}}(B)\Theta_{\text{pre}}(B^{12})\varepsilon_t, \qquad t \in \mathcal{T}_{\text{pre}}$$

$$\phi_{\mathrm{dur}}(B)\Phi_{\mathrm{dur}}(B^{12})(Y_t - \mu_{\mathrm{dur}}) = \theta_{\mathrm{dur}}(B)\Theta_{\mathrm{dur}}(B^{12})\varepsilon_t, \qquad t \in \mathcal{T}_{\mathrm{during}}$$

$$\phi_{\mathrm{post}}(B)\Phi_{\mathrm{post}}(B^{12})(Y_t - \mu_{\mathrm{post}}) = \theta_{\mathrm{post}}(B)\Theta_{\mathrm{post}}(B^{12})\varepsilon_t, \qquad t \in \mathcal{T}_{\mathrm{post}}$$

Because these models are fit separately, each model contains it's own intercept or drift term, ARMA parameters, and seasonal ARMA parameters, allowing more flexibility to account for the irregularities identified in the time plot of crime.

Goal: This is looking at the payoff between degrees of freedom and model complexity (see if there's any literature for this).

Part II SARIMAX with and without unemployment

In order to identify associations between crime and unemployment before, during, and after covid, unemployment is added as an exogenous variable. Using the Likelihood Ratio Test, we assess whether the addition of unemployment explains more variation in crime than covid and the ARMA and seasonal ARMA terms alone.

Goal: Identify whether unemployment is associated with crime.

Part III SARIMAX modeling unemployment with lag

Trend Considerations

The time plot in Figure 1 suggests that both a constant and linear trend are reasonable. The SARIMAX model with a drift would have the following equation

$$\phi(B)\Phi(B^{12})\left(Y_t - \mu - \delta t - \beta_1 D_{\mathrm{during},t} - \beta_2 D_{\mathrm{post},t}\right) = \theta(B)\Theta(B^{12})\varepsilon_t$$

In order to account for the possibility of drift, both SARIMAX with constant and linear trends were fit. Accounting for drift avoids non-stationarity concerns and differencing, allowing for the proper specification of the mean function with respect to time (double check. trying to say varies with time)

Model Selection Criteria

In order to restrict model checking (to models with…), the Akaike information criterion (AIC) was used as a means of comparison. Although AIC values provide guidance in comparing these models based on prediction, it should not be used as a statistical test (Lecture 5). Before considering the models further, the roots were checked for all models to ensure causality invertibility, and reducibility [Lecture 5]. Lastly, we required that the coefficients of the ARMA and seasonal ARMA terms be non-zero to avoid unnecessary model complexity.

2 Results

Initial data exploration revealed that time series of crime and unemployment had completely non-negative values. In the initial stages of testing models with both SARIMA and SARIMAX, numerical issues were present with near unit roots and